DiaTool-DPO

Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

5-State MDP · Automatic Pair Construction · Near GPT-4o Performance

ML Researchers & NLP Engineers | Tool-Augmented LLMs & RLHF Alignment

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

TA-LLMs trained only with SFT on expert trajectories struggle with real-world incomplete queries and out-of-scope requests, causing hallucinated tool invocations.

Failure Mode 1

Missing Slot-Filling

When user queries lack required arguments, SFT models skip clarifying questions and attempt tool calls with incomplete parameters.

Failure Mode 2

Hallucinated Tool Calls

Without preference feedback, models fabricate argument values rather than requesting missing information from the user.

Failure Mode 3

Unrejected Out-of-Scope

SFT models fail to reject tool calls when no suitable tool exists, producing invalid invocations for unsupported functionality.

Solution: DiaTool-DPO models TA-LLM dialogue as a 5-state MDP and uses DPO with automatically constructed trajectory pairs to teach preference between correct and incorrect dialogue flows.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

S1
Initial
No history; rejects out-of-scope requests here

S2
Tool Selected w/o Complete Slots
Tool identified; slot-filling QA repeats N times

S3
Tool Selected w/ Complete Slots
All args determined; ready to call

S4
Wait for Tool Response
Tool called; awaiting result

S5
Complete
Results received; user reply generated

reject → stay

slot-fill loop ×N

args complete

tool call

generate reply

S1 → S1 : Out-of-scope requests return a rejection and stay in Initial state (Type 3 query)

S2 loop : Slot-filling QA repeats N times (S1 → S2 → ... → S2 → S3) until all required parameters are gathered (Type 2 query)

Type 1 fast-path: S1 → S3 → S4 → S5 (all arguments supplied upfront; no slot-filling needed)

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Slot-Filling?	Example Scenario
Type 1	1 → 3 → 4 → 5	No – all args provided upfront	*"Book a flight from Seoul to Tokyo on May 1" (origin, dest, date all present)*
Type 2	1 → (2×N) → 3 → 4 → 5	Yes – N rounds of slot-filling QA	*"Book a flight" (missing origin, destination, date → model asks N clarifying questions)*
Type 3	1 → 1	N/A – request rejected	*"Write me a poem" (no suitable tool available → model returns rejection message)*

These three trajectories are the foundation for automatic preference pair construction: each 'chosen' trajectory is paired with a 'rejected' alternative that teaches a specific dialogue skill.

Automatic Paired Trajectory Construction — No Human Labels

Each DPO pair: Chosen (correct trajectory) ↔ Rejected (incorrect trajectory). Pairs teach three distinct lessons.

Query	Chosen →	Rejected →	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2×M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090 / 562	Easy / Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2×N)→3→4→5	1→(2×M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	–	1→1	Tool call accept	2,089 / 562	Easy / Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562	Hard

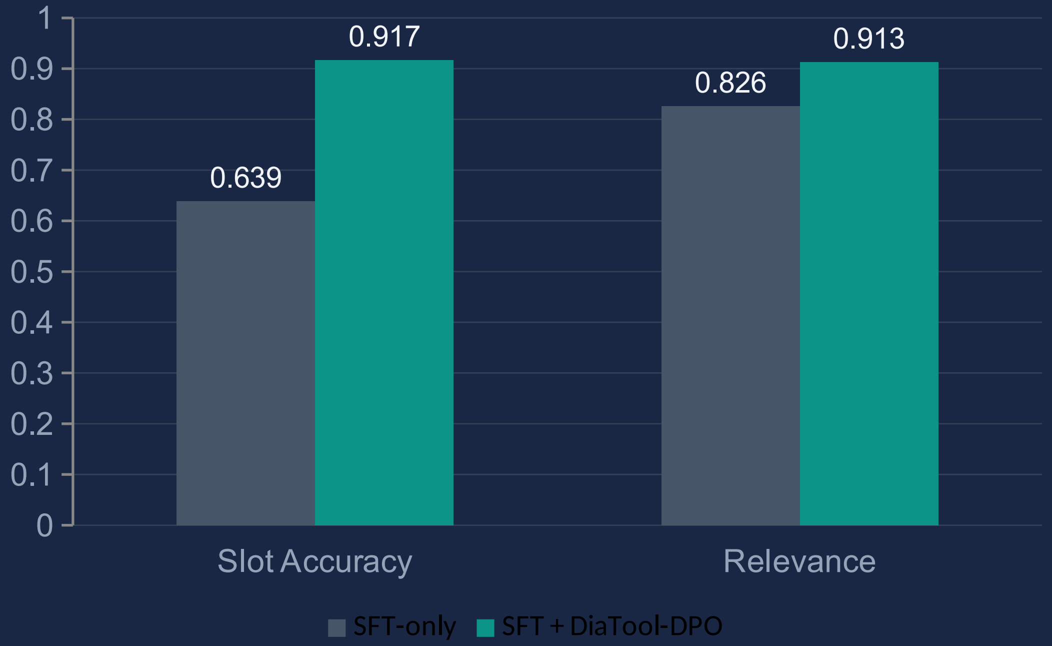
SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

FunctionChat-Bench evaluation — bold = best per model per metric

Model	Method	Call	Compl.	Slot	Relev.	Micro	Macro
Prop.-8B	DPO-only	0.314	0.700	**0.833**	0.609	0.575	0.614
	SFT-only	**0.900**	0.916	0.694	**0.913**	0.870	0.856
	SFT + DPO	0.886	**0.929**	**0.833**	0.826	**0.884**	**0.868**
Prop.-3.1B	DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
	SFT-only	**0.771**	0.817	0.750	**0.826**	**0.790**	0.791
	SFT + DPO	0.743	**0.871**	**0.833**	**0.826**	0.765	**0.818**
LLaMA-3-8B	DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
	SFT-only	0.843	**0.957**	0.639	0.826	0.844	0.816
	SFT + DPO ★	**0.857**	0.929	**0.917**	**0.913**	**0.905**	**0.904**

= Best Slot / Relevance for SFT+DPO ★ = LLaMA-3-8B SFT+DPO (largest absolute gains)

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO



Slot Accuracy

0.639 → **0.917**

+43.5% improvement

Relevance

0.826 → **0.913**

+10.5% improvement

Near GPT-4o performance: 94.8% information gathering, 91% tool call rejection (SFT + DiaTool-DPO, LLaMA-3-8B)

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

01 MDP Formulation Enables Systematic Pair Construction

Defining 5 internal states and 3 query types provides a principled structure to automatically generate chosen/rejected trajectory pairs — no human annotation required.

02 DPO Complements SFT — It Does Not Replace It

DiaTool-DPO-only performs poorly across all models. Optimal results require SFT first; DPO adds targeted preference signal on top of a strong supervised baseline.

03 Largest Gains in Slot-Filling and Tool Rejection

LLaMA-3-8B: Slot +43.5% (0.639 → 0.917), Relevance +10.5% (0.826 → 0.913). These are exactly the failure modes SFT alone cannot reliably address.

04 Language-Agnostic Method

Primary experiments in Korean generalize to English, opening broad applicability across language-diverse production TA-LLM deployments.

05 Path to GPT-4o-Level Conversational Tool Use on Smaller Models

SFT + DiaTool-DPO on LLaMA-3-8B reaches ~94.8% information gathering and ~91% tool call rejection — near GPT-4o performance at open-source model scale.